16/03/2022

NBTS

Corporate Office : Aakash Tower, 8, Pusa Road, New Delhi-110005, Phone : 011-47623456

MM : 720 **Time : 3.00 Hrs**

NCERT Booster Test Series

(for NEET-2022)

Test - 6

Topics covered :

Physics : Thermodynamics, Kinetic theory, Oscillations, Waves.

Chemistry : The p-Block Elements (Group-13 & 14), Organic Chemistry: Some Basic Principles and Techniques, Hydrocarbon, Environmental chemistry.

Botany : Transport in Plants, Mineral Nutrition, Photosynthesis in Higher Plants, Respiration in Plants, Plant Growth and Development.

Zoology : Animal Kingdom and Structural Organisation in Animals: Animal Morphology.

Instructions :

(i) There are two sections in each subject, i.e. Section-A & Section-B. You have to attempt all 35 questions from Section-A & only 10 questions from Section-B out of 15.

(ii) Each question carries 4 marks. For every wrong response 1 mark shall be deducted from the total score. Unanswered / unattempted questions will be given no marks.

(iii) Use blue/black ballpoint pen only to darken the appropriate circle.

(iv) Mark should be dark and completely fill the circle.

(v) Dark only one circle for each entry.

(vi) Dark the circle in the space provided only.

(vii) Rough work must not be done on the Answer sheet and do not use white-fluid or any other rubbing material on the Answer sheet.

PHYSICS

Choose the correct answer :

SECTION-A

1. Consider the following two statements

(A) No process is possible whose sole result is the transfer of heat from a colder object to a hotter object.

(B) A process can be reversible only if it is quasi-static (system in equilibrium with the surroundings at every stage) and there are no dissipative effects.

The correct statements(s) is/are

(1) Only (A) (2) Only (B)

(3) Both (A) and (B) (4) Neither (A) nor (B)

2. A heat engine works between the temperatures 400 K and 800 K. Which of the following can not be efficiency of the heat engine?

(1) 20% (2) 10%

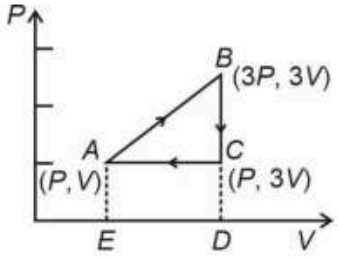
(3) 19% (4) 67%

3. The volume of an ideal monoatomic gas is decreased to 1/8 of its initial volume suddenly. The pressure increases by a factor of

(1) 8 (2) 16

(3) 32 (4) 7

4. An ideal gas is taken around *ABCA* thermodynamic cycle as shown in the following *P-V* diagram. The internal energy change during the complete cycle is



(1) 3*PV* (2) 6*PV*

(3) 8*PV* (4) Zero

5. Heat input for an ideal heat engine is 450 cal. If the efficiency of the engine is 45%, then the work done by the engine will be equal

(1) 16 cal (2) 20 cal

(3) 202.5 cal (4) 150 cal

6. A refrigerator operating on Carnot cycle is to maintain eatables kept inside at 6°C. If the outside temperature is 28°C. The coefficient of performance of the refrigerator will be

(1) 13.84 (2) 12.7

(3) 9.21 (4) 5.67

7. The average translational kinetic energy of ideal diatomic gas molecule, at temperature *T* Kelvin is equal to

(1) $\frac{3}{2} k_B T$ (2) $\frac{9}{2} k_B T$

(3) $\frac{7}{2} k_B T$ (4) $\frac{5}{2} k_B T$

8. The graph between the mean free path of ideal gas molecules as function of inverse of number density of molecules is

(1) Rectangular hyperbola

(2) Parabola

(3) Straight Line

(4) Ellipse

9. An air bubble of volume 1.0 cm^3 rises from the bottom of a lake 60 m deep at a temperature of 10°C. To what volume does it grow when it reaches the surface, which is a temperature of 25°C?

(1) 7.37 cm^3 (2) 2.89 cm^3

(3) 8.91 cm^3 (4) 5.12 cm^3

10. At what temperature is the rms speed of a molecule in a nitrogen gas cylinder equal to the rms speed of a helium gas atom at –20°C? (Molecular Mass of N_2 : 28 u)

(1) 1161°C (2) 1771°C

(3) 1498°C (4) 1000°C

11. If at a fixed pressure, the volume of a gas is proportional to its absolute temperature *T*. This law is called

(1) Boyle's Law (2) Charles' Law

(3) Avogadro Law (4) Dalton's Law

12. A geyser heats water flowing at the rate of 4 litres per minute from 30°C to 80°C. If the geyser operates on a gas burner, what is the rate of consumption of fuel if the heat of combustion of fuel is 5×10^5 J/g?

(1) 28 mg/s (2) 61 mg/s

(3) 69 g/s (4) 10 kg/s

13. What amount of heat must be supplied to 40 mg of nitrogen gas (at room temperature) to raise its temperature by 50°C at constant pressure? (Molecular mass of N_2 = 28 u)

(1) 3.89 J (2) 4.16 J

(3) 1.11 J (4) 2.08 J

14. The ratio of specific heat capacities of two copper solid spheres, of masses 2 kg and 3 kg respectively is

(1) 1:2 (2) 1:1

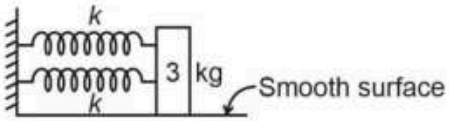
(3) 1:3 (4) 2:3

15. A body oscillates with SHM according to the equation $x = 4\sin\left(2\pi t + \frac{\pi}{4}\right)$. What is acceleration of the particle at t = 2.5 s? (displacement is in centimeter and time in seconds)

(1) 111.5 cm/s^2 (2) 132 cm/s^2

(3) 157.9 cm/s^2 (4) Zero

16. Two identical springs of spring constant 600 N/m each are attached to a block of mass 3 kg and to fixed support as shown in figure. What is its frequency of oscillations?



(1) $\frac{5}{\pi}$ Hz (2) $\frac{10}{\pi}$ Hz

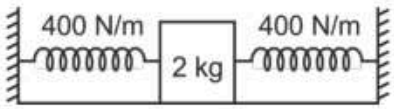
(3) $\frac{20}{\pi}$ Hz (4) $\frac{\pi}{2}$ Hz

17. A particle of mass m = 2 kg is following a force law; $F = -32x$, where x is displacement of particle from equilibrium position in meter and force F is in Newton. What is period of oscillation of the particle?

(1) 3.14 s (2) 6.28 s

(3) 1.57 s (4) 0.79 s

18. In the shown arrangement, the mass is displaced by 10 cm from equilibrium and released. The maximum speed of the block during simple harmonic motion will be



(1) 1 m/s (2) 2 m/s

(3) 4 m/s (4) 8 m/s

19. A particle executing simple hormonic motion has maximum velocity of 16 m/s and maximum acceleration of 64 m/s^2. What is its frequency of oscillations?

(1) $\frac{\pi}{2}$ Hz (2) $\frac{2}{\pi}$ Hz

(3) 2π Hz (4) π Hz

20. Which among the following examples does not represent nearly simple harmonic motion?

(1) Motion of a ball bearing inside a smooth concave bowl

(2) Oscillating mercury column in a vertical U-tube

(3) Simple pendulum

(4) A hydrogen molecule rotating about its centre of mass

21. Which of the following relationship between acceleration a and displacement x of a particle involves simple harmonic motion?

(1) $a = 0.5x$ (2) $a = -50x^3$

(3) $a = 100x^2$ (4) $a = -4x$

22. The motion of the particle executing simple harmonic motion is described by displacement equation $x = 6\sin(5\pi t + \phi)$. If the initial position of the particle (at $t = 0$) is 3 cm. What is initial phase of the particle?

(1) $\frac{\pi}{2}$ rad (2) $\frac{\pi}{3}$ rad

(3) $\frac{\pi}{6}$ rad (4) $\frac{\pi}{4}$ rad

23. A spring balance has a scale that reads from 0 to 100 kg. The length of scale is 10 cm. A block is suspended from this balance at free end. When displaced and released, oscillates with period of 0.32 sec. What is the approximate weight of the block?

(1) 3 kgf (2) 5 kgf

(3) 25 kgf (4) 6 kgf

24. The piston in a cylinder head of a locomotive has a stroke of 20 cm. The piston moves with simple harmonic motion with angular frequency of 5 rad/s. If piston mass is 500 g. What is maximum force acting on the piston?

(1) $\frac{5}{4}$ N (2) $\frac{7}{4}$ N

(3) $\frac{14}{9}$ N (4) $\frac{16}{3}$ N

25. A circular disc of mass 2 kg is suspended by a wire attached to its centre. The wire is twisted by rotating disc and released. The period of torsion oscillation is 3 s. The radius of disc is 50 cm. What is torsional spring constant of the wire?

(1) $3\pi^2$ Nm rad^{-1}

(2) $\frac{\pi^2}{9}$ Nm rad^{-1}

(3) $9\pi^2$ Nm rad^{-1}

(4) $6\pi^2$ Nm rad^{-1}

26. A simple pendulum is mounted in a cabin that is freely falling under gravity. If length of pendulum is L. Then its frequency in cabin will be

(1) $n = \frac{1}{2\pi}\sqrt{\frac{g}{L}}$ (2) $n = \frac{1}{\pi}\sqrt{\frac{g}{L}}$

(3) $n = 0$ (4) $n = \infty$

27. What is length of a simple pendulum on earth surface whose period is 1 second?

(1) 100 cm (2) 50 cm

(3) 25 cm (4) 40 cm

28. Given below are some examples of wave motion. Which among the given is incorrect?

(1) Waves produced by motor boat sailing in water are transverse and longitudinal wave.

(2) Waves produced in a cylinder containing liquid by moving a piston back and forth are longitudinal waves.

(3) Ultrasonic waves in air produced by a vibration quartz crystal are transverse waves.

(4) Motion of a kink in a longitudinal spring produced by displacing one end of a spring sideways is combination of longitudinal and transverse waves.

29. A wave travelling along a string is described by $y(x, t) = 3\sin(50x - 4t)$, numerical constants are in SI unit. With what velocity the wave is propagating?

(1) 3 cm/s (2) 4 cm/s

(3) 8 cm/s (4) 1.25 cm/s

30. The speed of transverse wave in a string is

(1) Proportional to tension in the string

(2) Proportional to radius of the string

(3) Proportional to length of the string

(4) Proportional to square root of tension in the string

31. The speed of longitudinal waves in solid bar of density ρ and Young's modulus Y is

(1) $v = \sqrt{Y\rho}$ (2) $v = \sqrt{\frac{Y}{\rho}}$

(3) $v = \frac{Y}{\sqrt{\rho}}$ (4) $v = \sqrt{Y\rho^2}$

32. The speed of sound in air at STP is $\left(\gamma_{air} = \frac{7}{5}\right)$

(1) 1600 m s^{-1} (2) 300 m s^{-1}

(3) 280 m s^{-1} (4) 331.3 m s^{-1}

33. A rocket is moving at a speed of 200 m/s towards a stationary target. While moving it emits waves of frequency 1000 Hz. Some waves are reflected back to rocket as echo. What is frequency of echo detected by the rocket? (velocity of sound = 330 m/s)

(1) 1500 Hz (2) 2540 Hz

(3) 3600 Hz (4) 4080 Hz

34. Two sound sources are producing sound waves of frequencies 415 Hz and 418 Hz. How many beats are produced per minute?

(1) 3 (2) 60

(3) 120 (4) 180

35. A string has mass of 25 gm uniformly distributed over its length of 20 m. The length is under a tension of 200 N. If a transverse jerk is struck at one end of the string, how long does the disturbance take to reach the other end?

(1) 0.02 s (2) 0.05 s

(3) 0.01 s (4) 0.10 s

SECTION-B

36. A transverse harmonic wave on a string is described by $y(x, t) = 4.0\sin\left(42t + 0.014x + \frac{\pi}{4}\right)$ where x and y are in cm and t in second. What is displacement of particle at $t = 0$ at origin?

(1) 4 cm (2) $2\sqrt{3}$ cm

(3) $\sqrt{2}$ cm (4) $2\sqrt{2}$ cm

37. The transverse displacement of a string clamped at both ends is given by $y(x, t) = 0.02\sin\left(\frac{2\pi}{3}x\right)\cos(100\pi t)$ where x and y are in meter and t in s. The length of string is 1 m and its mass is 20 g. What is the tension in the string?

(1) 300 N (2) 350 N

(3) 450 N (4) 600 N

38. A violin note and a sitar note may have same frequency yet we can distinguish between two notes. The characteristics of sound wave to distinguish is

(1) Loudness (2) Pitch

(3) Echo (4) Quality of sound

39. Using the formula $v = \sqrt{\frac{\gamma P}{\rho}}$, we can say that the speed of sound in air

(1) Changes with pressure

(2) Increases with increasing humidity

(3) Is independent of temperature

(4) Decreases with increasing humidity

40. A bat emits ultrasonic sound waves of frequency 500 kHz. The sound meets water surface. What is wavelength of transmitted sound. (speed of sound in air 360 m/s and in water 1500 m/s)

(1) 2 mm (2) 2.6 mm

(3) 3 mm (4) 7.2 mm

41. A particle is performing simple harmonic motion starting from extreme position. Graphical representation shows that between displacement and acceleration there is a phase difference of

(1) 0 rad (2) $\frac{\pi}{2}$ rad

(3) $\frac{\pi}{4}$ rad (4) π rad

42. The time period of a pendulum is T. If length of pendulum is made 4 times and mass is halved then new time period becomes

(1) $2T$ (2) $4T$

(3) T (4) $\frac{T}{2}$

43. The equation of damped simple harmonic motion is $\frac{d^2x}{dt^2} + \frac{b}{m}\frac{dx}{dt} + \frac{k}{m}x = 0$. Then angular frequency of oscillation is

(1) $\omega = \left(\frac{k}{m} - \frac{b^2}{4m^2}\right)^{\frac{1}{2}}$ (2) $\omega = \left(\frac{k}{m} - \frac{b}{2m}\right)^{\frac{1}{2}}$

(3) $\omega = \left(\frac{k}{m} - \frac{b^2}{4m}\right)^{\frac{1}{2}}$ (4) $\omega = \left(\frac{k}{m} - \frac{b}{2m^2}\right)^{\frac{1}{2}}$

44. The acceleration due to gravity on the surface of moon is 1.7 ms^{-2}. What is time period of simple pendulum on the surface of moon if its time period on earth surface is 1.5 second? (g_e = 9.8 $m\,s^{-2}$)

(1) 3.6 s (2) 2.4 s

(3) 1.5 s (4) 4.2 s

45. A cylindrical piece of cork of base area A and height L floats in liquid of density ρ. The cork is depressed slightly and then released. If density of cork is σ then time period of oscillation is

(1) $T = 2\pi\sqrt{\frac{L}{g} \times \frac{\sigma}{\rho}}$ (2) $T = 2\pi\sqrt{\frac{L}{g}}$

(3) $T = 2\pi\sqrt{\frac{L}{g} \times \frac{\rho}{\sigma}}$ (4) $T = 2\pi\sqrt{\frac{\sigma g}{\rho L}}$

46. Two sitar strings A and B playing a note, are slightly out of tune and produce 3 beats per second. Tension in B is slightly increased and beat frequency increases to 5 beats per second. What is original frequency of B if frequency of A is 317 Hz?

(1) 314 Hz (2) 320 Hz

(3) 317 Hz (4) 322 Hz

47. Consider the following statements and choose the correct option.

Statement 1: Doppler effect is a wave phenomenon.

Statement 2: Doppler's effect holds both for sound waves and electromagnetic waves.

(1) Statement 1 is true but statement 2 is false

(2) Statement 1 is false but statement 2 is true

(3) Both statements 1 and 2 are true

(4) Both statements 1 and 2 are false

48. Suppose a source of sound emitting frequency ν_0 is approaching a stationary observer, the speed of sound in air is v, then frequency heard by observer when ($v_s << v$) is

(1) $\nu_0\left(1 - \frac{v_s}{v}\right)$ (2) $\nu_0\left(1 + \frac{v_s}{v}\right)$

(3) $\nu_0\left(1 + \frac{v}{v_s}\right)$ (4) $\nu_0\left(1 - \frac{v}{v_s}\right)$

49. Keeping pressure of a gas constant and equal to 2 atm, the volume is changed from 2 litres to 1 litres. The work done on the gas will be equal to

(1) 200 J

(2) – 200 J

(3) 100 J

(4) 600 J

50. The product of an intensive variable and extensive variable is

(1) Intensive

(2) Extensive

(3) Either of (1) and (2) is possible

(4) None of these

CHEMISTRY

SECTION-A

51. Select the alkane having highest boiling point among given.

(1) Pentane (2) 2-methyl butane

(3) Hexane (4) Octane

52. Decarboxylation of sodium butanoate produces

(1) Methane (2) Butane

(3) Pentane (4) Propane

53. Kolbe's electrolysis of aqueous solution of potassium propanoate does not produce

(1) $CH_3CH_2CH_2CH_3$ (2) $H_2(g)$

(3) $CO_2(g)$ (4) $CH_3CH_2COOCH_3$

54. Select the correct statement about halogenation reactions of alkanes.

(1) Iodination of alkanes is too violent to be controlled

(2) These reactions are supposed to proceed via free radical chain mechanism

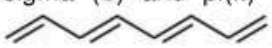
(3) These reactions involve heterolysis

(4) These reactions are example of addition reactions

55. Aromatization of hexane in the presence of Cr_2O_3 or V_2O_5 or Mo_2O_3 gives

(1) Styrene (2) Toluene

(3) Benzene (4) Mesitylene

56. Number of sigma (σ) and pi(π) bonds in the compound  respectively are

(1) 19, 4 (2) 19, 3

(3) 17, 4 (4) 18, 2

57. Which among the following compound does not exhibit geometrical isomerism?

(1) $(CH_3)_2C = CH_2$ (2) $ClCH = CHBr$

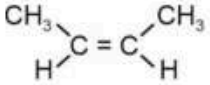
(3) $CH_3CH = CHCH_3$ (4) $CH_3CH = CHCl$

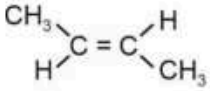
58. Major product obtained in the following reaction is

$$CH_3C \equiv CCH_3 + H_2 \xrightarrow{Na/Liquid\ NH_3}$$

(1) $CH_3CH_2CH_2CH_3$

(2) $CH_3CH_2CH_2CH_2CH_3$

(3) 

(4) 

59. Which among the following is the correct order of dehydrohalogenation of alkyl chlorides?

(1) Tert. > Secondary > Primary

(2) Primary > Secondary > Tert.

(3) Tert. > Primary > Secondary

(4) Secondary > Tert. > Primary

60. Consider the following reactions

(i) $CH_3CH = CHCH_3 \xrightarrow{KMnO_4/H^+}$ Product (A)

(ii) $CH_3CH = CH_2 \xrightarrow{cold\ conc.\ H_2SO_4}$ Product (B)

A and B respectively are

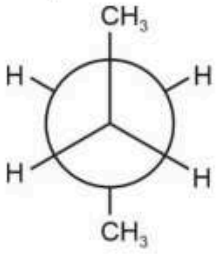
(1) CH_3COCH_3 and CH_3CHO

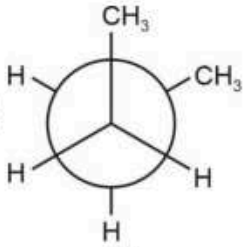
(2) CH_3COOH and CH_3CH_2OH

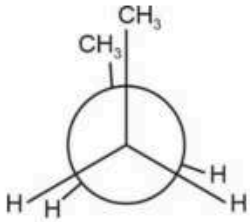
(3) CH_3COOH and $CH_3-\underset{\displaystyle OSO_2OH}{\underset{|}{CH}}-CH_3$

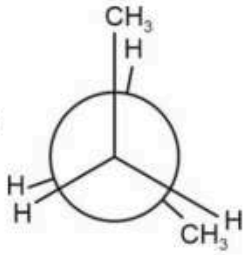
(4) CH_3CHO and $CH_3\underset{\displaystyle OSO_2OH}{\underset{|}{CH}}-CH_2-CH_3$

61. Most stable conformer of butane among the following

(1) 

(2) 

(3) 

(4) 

62. Which among the following species is aromatic in nature?

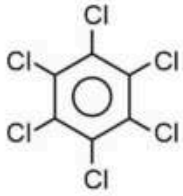
(1) 

(2) 

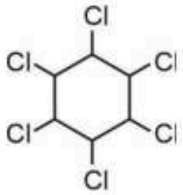
(3) 

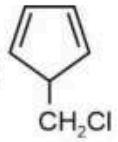
(4) 

63. When benzene is treated with excess of chlorine in the presence of anhyd $AlCl_3$, in cold, dark gives

(1) 

(2) 

(3) 

(4) 

64. Incorrect match of electrophile involved in given electrophilic substitution of benzene is

(1) Nitration : $\overset{+}{N}O_2$

(2) Chlorination : Cl^+

(3) Sulphonation: SO_2

(4) Friedel craft's methylation : $\overset{\oplus}{C}H_3$

65. Select the incorrect statement about polynuclear hydrocarbons containing more than two rings fused together , is

(1) These can be formed on incomplete combustion of tobacco, coal and petroleum

(2) These can damage DNA and causes cancer into human body

(3) All these are non-carcinogenic in nature

(4) 1,2-Benzpyrene is an example of polynuclear compound.

66. Reductive ozonolysis of 3,4-dimethylhept-3-ene produces

(1) Aldehydes only

(2) Ketones only

(3) Both aldehyde and ketone

(4) Esters only

67. Which among the following alkane does not obtained when CH_3CH_2Cl and $CH_3CH_2CH_2Cl$ are used as reactants in Wurtz reaction?

(1) Butane (2) Hexane

(3) Pentane (4) Heptane

68. Which among the following group is meta directing?

(1) $-COCH_3$ (2) $-NH_2$

(3) $-NHCOCH_3$ (4) $-OCH_3$

69. Consider the following statements

a. Troposphere extends upto the height of 10 km from sea level

b. Cloud formation occurs in troposphere region of atmosphere

c. Dust, mist, fumes etc pollutants are part of troposphere pollution

(1) a only (2) a and b only

(3) b and c only (4) a, b and c

70. The common component(s) of photochemical smog are

(1) Ozone and nitric oxide only

(2) Oxygen and PAN only

(3) Ozone, PAN and oxygen only

(4) Ozone, PAN, formaldehyde and acrolein

71. Maximum prescribed concentration of iron (in ppm) in drinking water is

(1) 0.2 (2) 20

(3) 200 (4) 2000

72. Select the **correct** statement among the following.

(1) Most herbicide are non-toxic to mammals

(2) Sodium arsenite is an example of pesticide

(3) Fuel obtained from plastic waste has very high octane rating

(4) 75% of the solar energy is radiated back to the earth and rest is absorbed by the earth surface

73. Earlier used solvent for dry cleaning was

(1) H_2O_2 (2) C_2Cl_6

(3) $Cl_2C = CCl_2$ (4) $CHCl_3$

74. Which among the following boron atom is not present?

(1) Borax (2) Kernite

(3) Orthoboric acid (4) Cryolite

75. Select the correct order of ionization enthalpy of 13 group elements, among given

(1) B > Al > Ga > In > Tl

(2) B > Tl > Ga > In > Al

(3) B > Tl > Ga > Al > In

(4) B > Ga > Tl > Al > In

76. Most stable oxidation state of thallium is

(1) +1 (2) +3

(3) +5 (4) –1

77. When aluminium dissolve in dil. HCl it gives

(1) $H_2(g)$ (2) Cl_2

(3) $O_2(g)$ (4) HOCl

78. Select the incorrect statement among the following.

(1) Boron cannot exceed its covalence more than four

(2) H_3BO_3 is a white crystalline solid

(3) H_3BO_3 is not soluble in hot water

(4) Acidic hydrolysis of $Na_2B_4O_7$, produces NaOH as one of the product

79. Select the reaction showing incorrect product

(1) $B_2H_6 + 3O_2 \rightarrow B_2O_3 + 3H_2O$

(2) $2BF_3 + 6NaH \rightarrow B_2H_6 + 6NaF$

(3) $4BF_3 + 3LiAlH_4 \rightarrow 2\ B_2H_6 + 3LiF + 3AlF_3$

(4) $B_2H_6 + 2NMe_3 \rightarrow [BH_2(NMe_3)_2] + [BH_4]^-$

80. Which among the following is least likely to exist?

(1) $[Sn(OH)_6]^{2-}$ (2) $[SiBr_6]^{2-}$

(3) $[GeCl_6]^{2-}$ (4) $[SiF_6]^{2-}$

81. Element which does not show catenation, is

(1) Carbon (2) Lead

(3) Germanium (4) Tin

82. Select the incorrect statement among the following

(1) Lead is unaffected by water

(2) SnF_4 is ionic in nature

(3) $SiCl_4$ does not hydrolyse

(4) $PbCl_2$ is more stable than $PbCl_4$

83. Buffer system which maintains the blood pH from 7.26 to 7.42, is a mixture of

(1) CO_2 and SO_2 (2) HCO_3^- and H_2CO_3

(3) HSO_4^- and HSO_3^- (4) CO and C_3O_2

84. When SiO_2 reacts with NaOH and HF separately it gives

(1) Na_2SiO_3 and NaF

(2) Na_2SiO_3 and SiF_4

(3) Na_2SiO_4 and NaF

(4) Na_2SiF_6 and $Si(OH)_4$

85. Which among the following statement is incorrect?

(1) The basic structural unit of silicates is SiO_4^{4-}

(2) Silicones are water repelling in nature

(3) Glass and cement are example of silicones

(4) Silicones can be used in surgical and cosmetic plants

SECTION-B

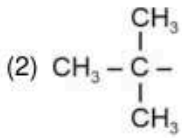
86. Molar mass (g mol^{-1}) of next homologous member of C_5H_{12} is

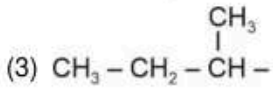
(1) 80 (2) 86

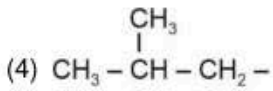
(3) 72 (4) 90

87. Sec. butyl groups among the following is

(1) $CH_3CH_2CH_2CH_2-$

(2) 

(3) 

(4) 

88. Which among the following functional group has highest priority during IUPAC naming?

(1) –CN (2) –CHO

(3) –COOR (4) –COOH

89. Which among the following group shows +R effect?

(1) –CN (2)

(3) – OCOR (4) $-NO_2$

90. Most stable carbocation among the following is

(1) $CH_3\overset{\oplus}{C}H_2$ (2) $(CH_3)_2\overset{\oplus}{C}H$

(3) $Ph\overset{\oplus}{C}H_2$ (4) $Ph\overset{\oplus}{C}(CH_3)_2$

91. In Carius method of estimation of halogen, 0.2 g of an organic compound gave 0.15 g of AgBr. Percentage of bromine in the compound will be (molar mass of AgBr is 188 g mol^{-1})

(1) 22% (2) 31.9%

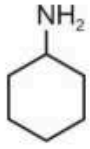
(3) 41.9% (4) 51.9%

92. Strongest acid among the following is

(1) CCl_3COOH (2) $CHCl_2COOH$

(3) $CH_2ClCOOH$ (4) CH_3COOH

93. Kjeldahl method is not applicable for

(1) 

(2) $C_6H_5NO_2$

(3) CH_3NH_2

(4) $(CH_3)_3N$

94. Metamerism can be shown by

(1) CH_3OCH_3 (2) $CH_3CH_2OCH_2CH_3$

(3) CH_3CH_2OH (4) $CH_3CH_2CH_2OH$

95. Homocyclic compound among the following is

(1) Furan (2) Thiophene

(3) Aniline (4) Pyridine

96. In the hydrocarbon

$\overset{7}{CH_3}-\overset{6}{C}\equiv\overset{5}{C}-\overset{4}{CH}=\overset{3}{C}=\overset{2}{CH}-\overset{1}{CH_3}$

The state of hybridization of carbons 3, 5 and 6 are in the following sequence

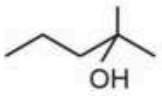
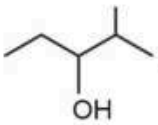
(1) *sp*, *sp*, sp^2 (2) *sp*, *sp*, *sp*

(3) *sp*, sp^2, *sp* (4) sp^2, sp^2, *sp*

97. Which of the following species is not electrophilic in nature?

(1) B_2H_6 (2) $\overset{\oplus}{N}H_4$

(3) $\overset{\oplus}{N}O_2$ (4) $\overset{+}{B}r$

98. Compounds  and  are related by

(1) Chain isomerism

(2) Position isomerism

(3) Functional isomerism

(4) Tautomerism

99. Separation of glycerol from spent-lye in soap industry is carried out by

(1) Simple distillation

(2) Fractional distillation

(3) Distillation under reduced pressure

(4) Steam distillation

100. Select the **incorrect** statement among the following.

(1) Column chromatography is an example of adsorption chromatography

(2) Thin layer chromatography plate is used in TLC

(3) Electromeric effect is temporary effect

(4) Colour of $[Fe(CN)_5NOS]^{4-}$ is blood red

BOTANY

SECTION-A

101. Who for the first time demonstrated that plants could be grown to maturity in a defined nutrient solution in complete absence of soil?

(1) Jan Ingenhousz (2) Julius von Sachs

(3) Joseph Priestley (4) T.W. Engelmann

102. Read the following statements and choose the option which is **true** for them.

Statement A: Deficiency of one element in plants can be met by supplying other elements.

Statement B: In the absence of essential elements plants do not complete their life cycle.

(1) Both the statements are correct

(2) Both the statements are incorrect

(3) Only statement A is correct

(4) Only statement B is correct

103. Match the following columns and select the **correct** option.

	Column I (Element)		**Column II** (Role in plant)
A.	Calcium	(i)	Maintenance of turgidity of cells
B.	Manganese	(ii)	Formation of mitotic spindle
C.	Potassium	(iii)	Formation of chlorophyll
D.	Iron	(iv)	Splitting of water during photosynthesis

(1) A(iii), B(i), C(iv), D(ii)

(2) A(i), B(iii), C(ii), D(iv)

(3) A(ii), B(iv), C(i), D(iii)

(4) A(iv), B(i), C(iii), D(ii)

104. Deficiency of which of the following groups of elements in plants causes necrosis, particularly of leaf tissues?

(1) N, Mn, Zn (2) N, S, Mo

(3) Ca, Cu, Mg (4) S, Fe, Mn

105. Identify the **true** statements from the following and select the option accordingly.

(a) Moderate decrease of micronutrients in plants causes toxicity.

(b) Manganese competes with magnesium for uptake in plants.

(c) In plants, induced deficiency of iron is due to excess of manganese.

(1) (b) only (2) (b) and (c)

(3) (a) and (c) (4) (c) only

106. The products of light reaction of photosynthesis are

(1) O_2, ATP and NADPH

(2) CO_2 and $NADH_2$

(3) ADP and NAD^+

(4) O_2 and $C_6H_{12}O_6$

107. In the chromatogram, chlorophyll *a* appears

(1) Yellow-green (2) Bright or blue-green

(3) Yellow (4) Yellow-orange

108. Out of the following pigments, which one shows highest peak of absorption spectrum between 400 – 500 nm wavelength of light?

(1) Carotenoid (2) Chlorophyll *a*

(3) Xanthophyll (4) Chlorophyll *b*

109. Select the **incorrect** statement w.r.t. photosynthetic units.

(1) Both photosystem I and II have light harvesting complex

(2) Each light harvesting complex has several reaction centres

(3) Light harvesting pigments are also called antenna molecules

(4) Each reaction centre has a single molecule of chlorophyll *a*

110. The final electron acceptor in non-cyclic photophosphorylation is

(1) Chlorophyll *a* (2) O_2

(3) $NADP^+$ (4) NAD^+

111. During photosynthesis, oxidation of water occurs
(1) In the stroma of chloroplast
(2) Towards lumen of thylakoid membrane
(3) Towards stroma side of inner membrane of chloroplast
(4) Between outer and inner membrane of chloroplast

112. Melvin Calvin studies on algal photosynthesis led to the discovery that the first CO_2 fixation product is a
(1) Three-carbon organic acid
(2) Five-carbon aldose sugar
(3) Four-carbon organic acid
(4) Five-carbon ketose sugar

113. The biosynthetic phase of photosynthesis called calvin cycle occurs in
(1) C_3 plants only
(2) Higher plants only
(3) All photosynthetic plants
(4) Thylakoid lumen of all green plants

114. The primary CO_2 acceptor molecule in the plants that show Kranz anatomy in the leaves is
(1) Phosphoenolpyruvate synthetase
(2) Ribulose bisphosphate
(3) 3-phosphoglycerate
(4) Phosphoenolpyruvate

115. The most abundant enzyme in the world has active site that can bind
(1) Both O_2 and CO_2 (2) Only CO_2
(3) Both N_2 and O_2 (4) Only O_2

116. Select the **incorrect** match from the following.
(1) Respiratory substrate – Carbohydrates
(2) Energy currency of the cell – ATP
(3) Gaseous exchange in plants – Lenticels
(4) Organism that can survive both anaerobically as well as aerobically – Obligate anaerobes

117. During glycolysis, two redox-equivalents in the form of two hydrogen atoms are removed
(1) Before the formation of glyceraldehyde-3-phosphate
(2) Before the first substrate level phosphorylation occurs
(3) Just after the second substrate level phosphorylation occurs
(4) When 3-phosphoglyceric acid is converted into 2- phosphoglycerate

118. During one complete citric acid cycle, how many CO_2 molecules are released?
(1) Four
(2) Three
(3) Two
(4) Six

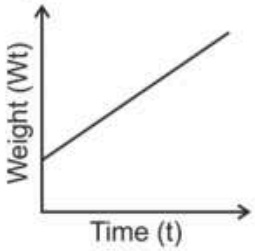
119. Net gain of ATP molecules directly through glycolysis and Krebs cycle from one molecule of glucose is
(1) Four (2) Six
(3) Thirty six (4) Thirty eight

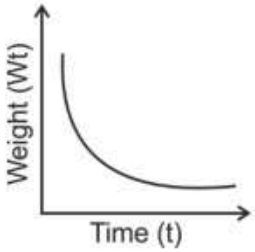
120. Glycerol can enter the respiratory pathway after being converted into
(1) Pyruvic acid
(2) Phosphoglyceraldehyde
(3) Acetyl CoA
(4) 3-phosphoglyceric acid

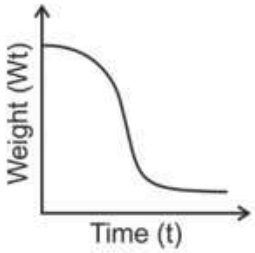
121. Growth in organisms is **not**
(1) A fundamental property
(2) Conspicuous
(3) A permanent increase
(4) Reversible

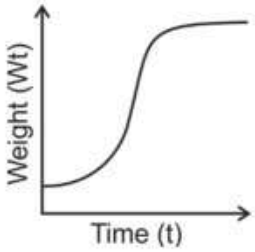
122. All of the following are the properties of the cells of meristematic region, **except**
(1) Thin cellulosic cell wall
(2) Rich in protoplasm
(3) Large sap vacuole
(4) Abundant plasmodesmatal connections

123. The graph that justify the growth expressed by the equation $W_t = W_0e^{rt}$ can be represented as

(1) 

(2) 

(3) 

(4) 

124. In coriander, the leaves of the juvenile plant are different in shape from those in mature plant. This is an example of

(1) Epinasty (2) Senescence

(3) Bolting (4) Plasticity

125. The plant growth regulator which is an adenine derivative is

(1) Ethylene (2) Kinetin

(3) Abscisic acid (4) Indole-3-acetic acid

126. A phytohormone used for thinning of fruits in cotton, cherry and walnut is

(1) A gaseous plant hormone

(2) Responsible for closure of stomata during water stress condition

(3) A derivative of carotenoid

(4) Alkaline in nature

127. Regarding seed dormancy, select the **odd** one out.

(1) Para-ascorbic acid

(2) Gibberellic acid

(3) Phenolic acid

(4) Immature embryo

128. Which one of the following microbes produces nitrogen fixing nodules on the roots of non-leguminous plant?

(1) *Azotobacter*

(2) *Frankia*

(3) *Nostoc*

(4) *Rhodospirillum*

129. Some plants respond to higher CO_2 concentration showing increased rates of photosynthesis leading to higher productivity. This is true for which of the following plants?

(1) Bell pepper

(2) Maize

(3) *Sorghum*

(4) Sugarcane

130. Read the following statements and choose the option which is true for them.

Statement 1: Diffusion is a slow process and is not dependent on a living system.

Statement 2: Diffusion is the only means for gaseous movement within the plant body.

(1) Both the statements are incorrect

(2) Both the statements are correct

(3) Only statement-1 is correct

(4) Only statement-2 is correct

131. Which of the following criteria pertains to active transport but not to both simple diffusion and facilitated transport?

(1) Highly selective in nature

(2) Transport saturates

(3) Uphill transport

(4) Requirement of special membrane proteins

132. Match the following columns and choose the **correct** option.

	Column I		Column II
A.	Guttation	(i)	Creates due to evaporative loss of water by plants
B.	Transpiration	(ii)	Occurs through stomata
C.	Root pressure	(iii)	Occurs through special openings of veins
D.	Transpiration pull	(iv)	A positive pressure inside the xylem

(1) A(ii), B(iv), C(i), D(iii)

(2) A(iv), B(iii), C(ii), D(i)

(3) A(iii), B(ii), C(iv), D(i)

(4) A(ii), B(iii), C(i), D(iv)

133. In plants, symplast pathway includes

a. Cytoplasm b. Intercellular space

c. Cell wall d. Plasma membrane

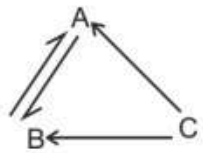
e. Plasmodesmata

The **correct** one(s) is/are

(1) a only (2) a, b, d and e

(3) b and c only (4) a, d and e only

134. The movement of water amongst three adjacent cells A, B and C is as follows



Find the **correct** statement that justifies this direction of movement of water.

(1) DPD of both the cell A and B is less than that of cell C

(2) Water potential of cell C is greater than that of cell A as well as cell B

(3) Water potential of cell A and B is more than that of cell C

(4) DPD of all the cells A, B and C are different from each other

135. The phenomenon in plants that maintains the shape and structure of the plants by keeping cells turgid is

(1) Photosynthesis

(2) Respiration

(3) Translocation of food

(4) Transpiration

SECTION-B

136. Ammonia is converted into nitrate with the help of

(1) *Pseudomonas* and *Thiobacillus*

(2) *Thiobacillus* and *Nitrobacter*

(3) *Pseudomonas* and *Nitrococcus*

(4) *Nitrosomonas* and *Nitrobacter*

137. Role of leg-haemoglobin in root nodules of legumes is to

(1) Create anaerobic conditions for nitrogenase

(2) Convert atmospheric nitrogen into ammonia

(3) Provide oxygen to *Rhizobium*

(4) Synthesize nitrogenase enzyme

138. According to chemiosmotic hypothesis, synthesis of ATP occurs when

(1) Electrons are passed from plastocyanin to photosystem I

(2) Protons enter into thylakoid lumen through plastoquinone

(3) Protons move from thylakoid lumen to stroma through transmembrane channel of CF_0

(4) Electrons are passed from ferredoxin to NADP reductase

139. For the synthesis of two hexose molecules, how many ATP and NADPH molecules respectively are used by C_3 plants?

(1) 30 and 12 (2) 24 and 12

(3) 36 and 24 (4) 60 and 24

140. The major limiting factor that influences the rate of photosynthesis is

(1) An external factor

(2) Availability of sunlight

(3) Availability of water in the soil

(4) Number of stomata in leaf

141. Which of the following molecules enter(s) the glycolytic pathway?

(a) Sucrose (b) Glucose

(c) Cellulose (d) Fructose

(1) (b) only (2) (a), (b) and (d)

(3) (b) and (d) only (4) (a) and (c)

142. Match the following columns and select the **correct** option w.r.t. ETS of mitochondria.

Column I	**Column II**
a. Succinate dehydrogenase	(i) Contains two copper centres
b. Cytochrome *c* oxidase complex	(ii) ATP synthase
c. Cytochrome *c*	(iii) Attached to the outer surface of inner mitochondrial membrane
d. Complex V	(iv) Provides reducing equivalents to ubiquinone

(1) a(ii), b(iv), c(iii), d(i) (2) a(iii), b(iv), c(i), d(ii)

(3) a(iv), b(ii), c(i), d(iii) (4) a(iv), b(i), c(iii), d(ii)

143. When $C_2H_2O_4$ is used as respiratory substrate and is completely oxidised, the respiratory quotient will be

(1) 4 (2) 1

(3) 0.7 (4) 0.9

144. Select the **incorrect** match from the following.

(1) Meristematic tissue – Cambium

(2) Dedifferentiated tissue – Cork

(3) Redifferentiated tissue – Secondary xylem

(4) Differentiated tissue – Collenchyma

145. Which of the following is **not** a function of plant growth regulator that was first isolated from corn-kernels and coconut milk?

(1) Promotion of female flowers in monoecious plants

(2) Production of chloroplast in leaves

(3) Growth of lateral shoot

(4) Overcome the apical dominance

146. The reducing agent NADH + H^+ is oxidised during

(1) Glycolysis

(2) Oxidative decarboxylation of pyruvic acid

(3) Lactic acid and alcohol fermentation

(4) Citric acid cycle

147. Spraying juvenile conifers with _____ hastens the maturity period and thus leading to early seed production.

Select the **correct** option to fill in the blank.

(1) Cytokinins

(2) Auxins

(3) Ethylene

(4) Gibberellins

148. Porins are proteins that form large pores in

(1) Outer membrane of mitochondria

(2) Cell wall of plants

(3) Inner nuclear membrane

(4) Cell wall of bacteria

149. Transpiration is affected by several external factors. Which of the following is **not** amongst them?

(1) Canopy structure

(2) Temperature

(3) Wind speed

(4) Humidity

150. The girdling experiment that was used to identify the tissues through which food is transported shows that

(1) Phloem is responsible for translocation of food in both the directions

(2) Translocation of food occurs through phloem in one direction

(3) Food is translocated by both xylem and phloem only in one direction

(4) Food is translocated by xylem and phloem in both the directions

ZOOLOGY

SECTION-A

151. The common feature that is exhibited by the organisms belonging to phylum Ctenophora and Coelenterata is

(1) Presence of cnidoblasts

(2) Polyp body form

(3) Exclusively marine

(4) Extracellular and intracellular digestion

152. Match the zoological name of animals listed in column 'A' with their common name in column 'B' and select the correct option.

	Column A		**Column B**
a.	*Pristis*	(I)	Flying fox
b.	*Pteropus*	(II)	Peacock
c.	*Psittacula*	(III)	Sawfish
d.	*Pavo*	(IV)	Parrot

(1) a-I, b-II, c-III, d-IV

(2) a-IV, b-III, c-II, d-I

(3) a-III, b-I, c-IV, d-II

(4) a-III, b-I, c-II, d-IV

153. The animals in which mesoderm is present as scattered pouches in between the ectoderm and endoderm are

(1) *Anopheles* and *Pila*

(2) *Wuchereria* and *Ancylostoma*

(3) *Ascaris* and *Taenia*

(4) *Pleurobrachia* and *Ctenoplana*

154. *Columba* and *Periplaneta* are similar in having

(1) Malpighian tubules in hindgut

(2) Air sacs connected to lungs

(3) Ventral heart

(4) Additional chambers in digestive tract

155. Read the given statements carefully and choose the correct one w.r.t. nocturnal omnivore belonging to the largest phylum of Animalia.

(1) In each segment, endoskeleton has hardened plates called sclerites that are joined to each other by a membrane.

(2) Each Malpighian tubule is lined by squamous epithelium

(3) Each eye consists of about 2000 tetragonal ommatidia

(4) They show paurometabolous development in which nymphs look very much like adults.

156. The pathway of water transport that is helpful in food gathering, gaseous exchange and removal of waste is characteristically seen in

(1) *Sepia* (2) *Sycon*

(3) Salpa (4) Struthio

157. Read the statements given below carefully w.r.t roundworms.

A. Found only in aquatic habitat or as parasites in animals.

B Male and female sex organs are present in separate roundworms.

C. Males and females are similar in length and width.

D. Fertilisation is internal.

Select the option that contains **incorrect** statements only

(1) A and B (2) B and C

(3) B and D (4) A and C

158. The vertebrates which are devoid of paired fins or limbs are

(1) *Ornithorhynchus* and *Petromyzon*

(2) *Salamandra* and *Scoliodon*

(3) *Calotes* and *Catla*

(4) *Myxine* and *Bungarus*

159. How many organisms from the box given below have both open circulatory system and true coelom?

Nereis, *Bombyx*, *Balanoglossus*, *Gorgonia*, *Chaetopleura*, *Wuchereria*

(1) Five (2) Three

(3) Four (4) Two

160. The pair of wings in cockroach that are opaque, dark and leathery, arise from _____ and are called _____. Choose the option that correctly fills the blanks respectively.

(1) Prothorax, Forewings

(2) Mesothorax, Hindwing

(3) Mesothorax, Tegmina

(4) Metathorax, Forewings

161. The ocellus in arthropods is a type of

(1) Thermoreceptor (2) Photoreceptor

(3) Mechanoreceptor (4) Osmoreceptor

162. Choose the **incorrect** option w.r.t fertilised eggs encased in capsules in cockroach.

(1) Dark reddish to blackish brown capsule

(2) About 8 mm long capsule

(3) Dropped or glued in a crack or crevice of low relative humidity

(4) Dropped usually near a food source

163. Choose the correct option regarding the given statements A and B.

Statement A : Water canal system is the most distinctive feature of echinoderms.

Statement B : A complete digestive system is present in *Asterias.*

(1) Both statements A and B are correct.

(2) Both statements A and B are incorrect.

(3) Only statement A is correct.

(4) Only statement B is correct.

164. How many of the following animals given in the box are not exclusively marine?

Ophiura, *Saccoglossus*, Hydra, *Spongilla* *Ctenoplana*, *Antedon*

(1) Two (2) Three

(3) Four (4) Five

165. Identify the figure given below and select the **incorrect** answer w.r.t it.



(1) Larvae migrates to fresh water for spawning.

(2) Devoid of scales

(3) Sucking type of mouth without jaws

(4) Cartilaginous skeleton

166. The common name of *Trygon* is ___ while that of *Exocoetus* is ___ respectively.

Select the **correct** option to fill in the blanks.

(1) Sting ray, fighting fish

(2) Electric ray, flying fish

(3) Sting ray, flying fish

(4) Electric ray, fighting fish

167. Choose the **incorrect** option w.r.t the phylum containing feather like gills.

(1) Anterior head has sensory tentacles

(2) All members are aquatic

(3) Body is unsegmented

(4) Mouth contains a file-like rasping organ

168. *Laccifer* and *Octopus* show similarity with *Echinus* in all the given options **except**

(1) Complete digestive tract

(2) Dioecious

(3) Triploblastic

(4) External fertilization

169. Appearance of a third germinal layer was first time seen in phylum 'X'. Select the animal that belongs to the same phylum 'X'.

(1) *Ctenoplana* (2) *Pennatula*

(3) *Planaria* (4) *Hirudinaria*

170. Mouth of Sea horse is __X__ whereas the mouth of Dog fish is located __Y__ . Select the correct option to fill the blanks.

	X	**Y**
(1)	Terminal	Ventrally
(2)	Ventral	Laterally
(3)	Dorsal	Terminally
(4)	Lateral	Dorsally

171. Body of the members of phylum to which Portuguese man-of-war belongs, can be divided into two identical halves in/by

(1) Only one plane passing through the horizontal axis of the body

(2) Any plane passing through the central axis of the body

(3) No plane

(4) Only one plane passing through longitudinal axis of the body

172. Complete the analogy and select the **correct** option.

Human : Alveoli : : Cockroach

(1) Spiracles (2) Tracheae

(3) Hemolymph (4) Tracheoles

173. Given below are two series A and B w.r.t cockroach.

Series A: Hepatic caeca, Malpighian tubules, nephrocytes, uricose glands.

Series B: Antennae, maxillary palps, labrum, labial palps, anal cerci.

Choose the **correct** option which includes one odd structure of both series 'A' and 'B'.

	Series A	**Series B**
(1)	Nephrocytes	Maxillary palps
(2)	Uricose glands	Antennae
(3)	Malpighian tubules	Labial palps
(4)	Hepatic caeca	Anal cerci

174. The approximate number of eggs contained in 5 oothecae of a cockroach is

(1) 80 (2) 60

(3) 45 (4) 16

175. Genital pouch in male cockroach is bounded by

(1) 9th – 10th sternum dorsally and 9th tergum ventrally

(2) 7th – 10th sternum ventrally and 10th tergum dorsally

(3) 9th – 10th terga dorsally and 9th sternum ventrally

(4) 7th – 8th terga dorsally and 7th sternum ventrally

176. **Assertion (A):** Cockroach has mosaic vision with less sensitivity but more resolution.

Reason (R): Each eye of cockroach has several ommatidia with the help of which it can receive several images of an object.

In the light of the above statements, choose the correct answers from the options given below.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

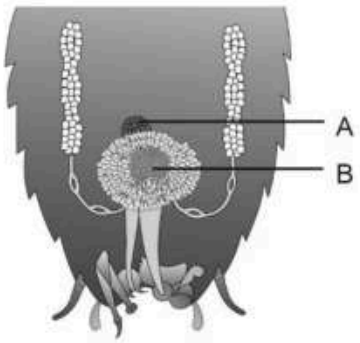
(4) (A) is false but (R) is true

177. Select the odd one among the given chordates w.r.t endoskeleton.

(1) *Petromyzon* (2) *Scoliodon*

(3) *Myxine* (4) *Clarias*

178. Following is a diagrammatic representation of male reproductive system of cockroach.



Select the option representing correct labelling and function performed by structures labelled as 'A' and 'B'.

(1) A – Ejaculatory duct – transport sperm
B – Phallomere – helps in copulation

(2) A – Spermatheca – Store sperms
B – Collaterial glands – Nourishes sperms

(3) A – Phallic gland – secretes the outer layer of spermatophore
B – Mushroom gland – Nourishes sperms

(4) A – Testis – Formation of sperms
B – Spermatophores – helps in copulation

179. Read the following statements and choose the option that correctly states them as True (T) or False (F).

A. Cockroach is able to move its head in all directions.

B. The space between mantle and hump in molluscs is called muscular foot.

C. In the gardens, earthworm can be traced by their faecal deposits.

D. *Aedes* is a gregarious pest.

	A	B	C	D
(1)	T	T	F	F
(2)	T	F	T	T
(3)	T	F	T	F
(4)	F	F	T	T

180. All of the given statements are correct w.r.t alimentary canal of *Periplaneta americana* **except**

(1) Gizzard helps in grinding of the food particles

(2) The entire midgut is lined by cuticle

(3) The hindgut is broader than midgut

(4) Crop is a sac like structure used for storing of food

181. Which of the following is **correct** for the arthrodial membrane?

(1) Thin and non-flexible

(2) Thick and non-flexible

(3) Thin and flexible

(4) Thick and flexible

182. One example of animal having a single opening to outside that serves the function of both, mouth as well as anus is

(1) *Salpa* (2) *Pennatula*

(3) *Apis* (4) *Cucumaria*

183. The *Corvus* can be differentiated from *Chelone* on the basis of

(1) Presence or absence of notochord

(2) Ventral nerve cord

(3) Chambers in heart

(4) Type of fertilisation

184. Select the taxon which contains both marine and fresh water species.

(1) Ctenophora (2) Annelida

(3) Echinodermata (4) Hemichordata

185. The excretory structure in phylum whose members are mostly endoparasites in humans and have dorsoventrally flattened body is

(1) Proboscis gland (2) Malpighian tubules

(3) Flame cells (4) Green glands

SECTION – B

186. The name of class 'X' refers to their creeping mode of locomotion. Members of this class differs from the members of preceding class in

(1) Tympanum representing ear

(2) Oviparity

(3) Lack of capacity to regulate their body temperature

(4) Type of development

187. During summer, *Rana tigrina* takes shelter in deep burrows to protect them from extreme heat. This protective mechanism is termed as

(1) Hibernation (2) Aestivation

(3) Mimicry (4) Camouflage

188. Which of the following is an **incorrect** pair for the earthworm?

(1) Female genital pore – 14th segment

(2) Clitellum – $14\text{-}16^{th}$ segment

(3) Male genital pore – 14th segment

(4) Stomach – $9\text{-}14^{th}$ segment

189. Select the **correct** flow of deoxygenated blood in the heart of *Rana tigrina*.

(1) Vena cava → right atrium → sinus venosus → ventricle → conus arteriosus

(2) Vena cava → sinus venosus → right atrium → ventricle → conus arteriosus

(3) Vena cava → sinus venosus → left atrium → left ventricle → pulmonary artery

(4) Pulmonary vein → left atrium → ventricle → conus arteriosus

190. Digestive system of frog differs from humans in

(1) Secretion of liver

(2) Finger like folds in intestine

(3) Length of intestine

(4) Function of oesophagus

191. Which of the following is **incorrect** for the nephridia that are present from segment 3 to the last of earthworm?

(1) They are of three types

(2) Abundant in typhlosole

(3) They regulate the volume and composition of body fluids

(4) Some of them open on the surface of body wall

192. Match Column I with Column II and select the correct option.

	Column I		**Column II**
a.	*Petromyzon*	(i)	Body with scales
b.	*Branchiostoma*	(ii)	Ectoparasite on fishes
c.	*Doliolum*	(iii)	Notochord persistent throughout life
d.	*Hippocampus*	(iv)	Notochord present only in larval tail

	a	b	c	d
(1)	(ii)	(iii)	(iv)	(i)
(2)	(ii)	(iv)	(iii)	(i)
(3)	(ii)	(iii)	(i)	(iv)
(4)	(i)	(ii)	(iii)	(iv)

193. Read the given statements and choose **correct** option.

Statement A : Central nervous system of non-chordates is ventral, solid and single.

Statement B : Eyes of *Rana tigrina* are bulged and covered by a nictitating membrane to protect them in water.

(1) Only statement A is correct

(2) Only statement B is correct

(3) Both statements A and B are correct

(4) Both statements A and B are incorrect

194. Following table is a comparison of few characteristics of *Pheretima* and *Culex*. Select the **incorrect** one.

	Characteristics	***Pheretima***	***Culex***
(1)	Metameric segmentation	Yes	Yes
(2)	Circulatory system	Open	Open
(3)	Dioecious	No	Yes
(4)	Paired ganglia with double ventral nerve cord	Yes	Yes

195. All of the given traits are common between *Aptenodytes* and *Ornithorhynchus* **except**

(1) Homoiothermy

(2) Toothless jaw

(3) Oviparity

(4) Preen gland

196. The type of epithelium lining the Malpighian tubules of cockroach is

(1) Non-glandular and non-ciliated

(2) Non-glandular and ciliated

(3) Glandular and ciliated

(4) Glandular and non-ciliated

197. The presence of vocal sacs and a copulatory pad on the first digit of forelimbs is a distinguishable feature between male and females of organism 'X' belonging to class 'Y' of phylum Chordata.

Select the **correct** answer for X and Y.

	(X)	**(Y)**
(1)	*Naja*	Reptilia
(2)	*Rana*	Amphibia
(3)	*Betta*	Osteichthyes
(4)	*Neophron*	Aves

198. Choose the **incorrect** match w.r.t animal that is also called farmers friend.

	A	B	C
(1)	Testis	2 pairs	10th and 11th segments
(2)	Accessory glands	2 pairs	17th and 19th segments
(3)	Spermathecae	4 pairs	6th–9th segments
(4)	Ovary	2 pairs	12th and 13th segments

199. Select the correct set that contains economically important animals only.

A. *Rana* B. *Pheretima*

C. *Nereis* D. *Locusta*

(1) A and B (2) A and C

(3) B and D (4) C Only

200. Read the statements given and select the option that correctly identifies them as True(T) or False(F).

A. *Hyla* presents with two auricles and one ventricle in the heart

B. *Naja* and *Bungarus* are not poisonous

C. Platypus is a viviparous mammal

	A	B	C
(1)	T	T	F
(2)	T	T	T
(3)	F	F	T
(4)	T	F	F

❑ ❑ ❑

16/03/2022

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Test – 6

Answer key

1. (3)	41. (4)	81. (2)	121. (4)	161. (2)
2. (4)	42. (1)	82. (3)	122. (3)	162. (3)
3. (3)	43. (1)	83. (2)	123. (4)	163. (4)
4. (4)	44. (1)	84. (2)	124. (4)	164. (1)
5. (3)	45. (1)	85. (3)	125. (2)	165. (1)
6. (2)	46. (2)	86. (2)	126. (1)	166. (3)
7. (1)	47. (3)	87. (3)	127. (2)	167. (2)
8. (3)	48. (2)	88. (4)	128. (2)	168. (4)
9. (1)	49. (1)	89. (3)	129. (1)	169. (3)
10. (3)	50. (2)	90. (4)	130. (2)	170. (1)
11. (2)	51. (4)	91. (2)	131. (3)	171. (2)
12. (1)	52. (4)	92. (1)	132. (3)	172. (4)
13. (4)	53. (4)	93. (2)	133. (4)	173. (4)
14. (2)	54. (2)	94. (2)	134. (2)	174. (1)
15. (1)	55. (3)	95. (3)	135. (4)	175. (3)
16. (2)	56. (3)	96. (2)	136. (4)	176. (4)
17. (3)	57. (1)	97. (2)	137. (1)	177. (4)
18. (2)	58. (4)	98. (2)	138. (3)	178. (3)
19. (2)	59. (1)	99. (3)	139. (3)	179. (3)
20. (4)	60. (3)	100. (4)	140. (1)	180. (2)
21. (4)	61. (1)	101. (2)	141. (3)	181. (3)
22. (3)	62. (3)	102. (4)	142. (4)	182. (2)
23. (3)	63. (1)	103. (3)	143. (1)	183. (3)
24. (1)	64. (3)	104. (3)	144. (2)	184. (2)
25. (2)	65. (3)	105. (2)	145. (1)	185. (3)
26. (3)	66. (2)	106. (1)	146. (3)	186. (4)
27. (3)	67. (4)	107. (2)	147. (4)	187. (2)
28. (3)	68. (1)	108. (4)	148. (1)	188. (3)
29. (3)	69. (4)	109. (2)	149. (1)	189. (2)
30. (4)	70. (4)	110. (3)	150. (2)	190. (3)
31. (2)	71. (1)	111. (2)	151. (4)	191. (2)
32. (4)	72. (3)	112. (1)	152. (3)	192. (1)
33. (4)	73. (3)	113. (3)	153. (2)	193. (2)
34. (4)	74. (4)	114. (4)	154. (4)	194. (2)
35. (2)	75. (3)	115. (1)	155. (4)	195. (4)
36. (4)	76. (1)	116. (4)	156. (2)	196. (3)
37. (3)	77. (1)	117. (2)	157. (4)	197. (2)
38. (4)	78. (3)	118. (3)	158. (4)	198. (4)
39. (2)	79. (4)	119. (1)	159. (2)	199. (3)
40. (3)	80. (2)	120. (2)	160. (3)	200. (4)

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Hints and Solutions

PHYSICS

1. Answer (3)
NCERT Reference: Class XI, Part-I, Page No. 315
2. Answer (4)
NCERT Reference: Class XI, Part-I, Page No. 317
3. Answer (3)
NCERT Reference: Class XI, Part-I, Page No. 312
4. Answer (4)
NCERT Reference: Class XI, Part-I, Page No. 312
5. Answer (3)
NCERT Reference: Class XI, Part-I, Page No. 313
6. Answer (2)
NCERT Reference: Class XI, Part-I, Page No. 313
7. Answer (1)
NCERT Reference: Class XI, Part-I, Page No. 334
8. Answer (3)
NCERT Reference: Class XI, Part-I, Page No. 336
9. Answer (1)
NCERT Reference: Class XI, Part-I, Page No. 339
10. Answer (3)
NCERT Reference: Class XI, Part-I, Page No. 339
11. Answer (2)
NCERT Reference: Class XI, Part-I, Page No. 326
12. Answer (1)
NCERT Reference: Class XI, Part-I, Page No. 321
13. Answer (4)
NCERT Reference: Class XI, Part-I, Page No. 321
14. Answer (2)
NCERT Reference: Class XI, Part-I, Page No. 308
15. Answer (1)
NCERT Reference: Class XI, Part-I, Page No. 349
16. Answer (2)
NCERT Reference: Class XI, Part-I, Page No. 350
17. Answer (3)
NCERT Reference: Class XI, Part-I, Page No. 349
18. Answer (2)
NCERT Reference: Class XI, Part-I, Page No. 349
19. Answer (2)
NCERT Reference: Class XI, Part-I, Page No. 360

20. Answer (4)

NCERT Reference: Class XI, Part-I, Page No. 362

21. Answer (4)

NCERT Reference: Class XI, Part-I, Page No. No. 363

22. Answer (3)

NCERT Reference: Class XI, Part-I, Page No. 364

23. Answer (3)

NCERT Reference: Class XI, Part-I, Page No. 364

24. Answer (1)

NCERT Reference: Class XI, Part-I, Page No. 365

25. Answer (2)

NCERT Reference: Class XI, Part-I, Page No. 366

26. Answer (3)

NCERT Reference: Class XI, Part-I, Page No. 365

27. Answer (3)

NCERT Reference: Class XI, Part-I, Page No. 355

28. Answer (3)

NCERT Reference: Class XI, Part-I, Page No. 370

29. Answer (3)

NCERT Reference: Class XI, Part-I, Page No. 373

30. Answer (4)

NCERT Reference: Class XI, Part-I, Page No. 374

31. Answer (2)

NCERT Reference: Class XI, Part-I, Page No. 375

32. Answer (4)

NCERT Reference: Class XI, Part-I, Page No. 376

33. Answer (4)

NCERT Reference: Class XI, Part-I, Page No. 387

34. Answer (4)

NCERT Reference: Class XI, Part-I, Page No. 389

35. Answer (2)

NCERT Reference: Class XI, Part-I, Page No. 391

36. Answer (4)

NCERT Reference: Class XI, Part-I, Page No. 392

37. Answer (3)

NCERT Reference: Class XI, Part-I, Page No. 392

38. Answer (4)

NCERT Reference: Class XI, Part-I, Page No. 393

39. Answer (2)

NCERT Reference: Class XI, Part-I, Page No. 391

40. Answer (3)

NCERT Reference: Class XI, Part-I, Page No. 391

41. Answer (4)

NCERT Reference: Class XI, Part-I, Page No. 345

42. Answer (1)

NCERT Reference: Class XI, Part-I, Page No. 351

43. Answer (1)

NCERT Reference: Class XI, Part-I, Page No. 356

44. Answer (1)

NCERT Reference: Class XI, Part-I, Page No. 361

45. Answer (1)

NCERT Reference: Class XI, Part-I, Page No. 361

46. Answer (2)

NCERT Reference: Class XI, Part-I, Page No. 384

47. Answer (3)

NCERT Reference: Class XI, Part-I, Page No. 384

48. Answer (2)

NCERT Reference: Class XI, Part-I, Page No. 385

49. Answer (1)

NCERT Reference: Class XI, Part-I, Page No. 312

50. Answer (2)

NCERT Reference: Class XI, Part-I, Page No. 310

CHEMISTRY

51. Answer (4)

NCERT Reference: Class XI, Part-I, Page No. 380

52. Answer (4)

NCERT Reference: Class XI, Part-I, Page No. 379

53. Answer (4)

NCERT Reference: Class XI, Part-I, Page No. 379

54. Answer (2)

NCERT Reference: Class XI, Part-II, Page No. 381

55. Answer (3)

NCERT Reference: Class XI, Part-II, Page No. 382

56. Answer (3)

NCERT Reference: Class XI, Part-II, Page No. 385

57. Answer (1)

NCERT Reference: Class XI, Part-II, Page No. 387

58. Answer (4)

NCERT Reference: Class XI, Part-II, Page No. 387

59. Answer (1)

NCERT Reference: Class XI, Part-II, Page No. 388

60. Answer (3)

NCERT Reference: Class XI, Part-II, Page No. 391

61. Answer (1)

NCERT Reference: Class XI, Part-II, Page No. 383

62. Answer (3)

NCERT Reference: Class XI, Part-II, Page No. 399

63. Answer (1)

NCERT Reference: Class XI, Part-II, Page No. 401

64. Answer (3)

NCERT Reference: Class XI, Part-II, Page No. 400

65. Answer (3)

NCERT Reference: Class XI, Part-II, Page No. 403

66. Answer (2)

NCERT Reference: Class XI, Part-II, Page No. 401

67. Answer (4)

NCERT Reference: Class XI, Part-II, Page No. 405

68. Answer (1)

NCERT Reference: Class XI, Part-II, Page No. 403

69. Answer (4)

NCERT Reference: Class XI, Part-II, Page No. 407

70. Answer (4)

NCERT Reference: Class XI, Part-II, Page No. 412

71. Answer (1)

NCERT Reference: Class XI, Part-II, Page No. 416

72. Answer (3)

NCERT Reference: Class XI, Part-II, Page No. 428

73. Answer (3)

NCERT Reference: Class XI, Part-II, Page No. 420

74. Answer (4)

NCERT Reference: Class XI, Part-II, Page No. 317

75. Answer (3)
NCERT Reference: Class XI, Part-II, Page No. 318

76. Answer (1)
NCERT Reference: Class XI, Part-II, Page No. 318

77. Answer (1)
NCERT Reference: Class XI, Part-II, Page No. 319

78. Answer (3)
NCERT Reference: Class XI, Part-II, Page No. 320

79. Answer (4)
NCERT Reference: Class XI, Part-II, Page No. 321

80. Answer (2)
NCERT Reference: Class XI, Part-II, Page No. 324

81. Answer (2)
NCERT Reference: Class XI, Part-II, Page No. 235

82. Answer (3)
NCERT Reference: Class XI, Part-II, Page No. 324

83. Answer (2)
NCERT Reference: Class XI, Part-II, Page No. 328

84. Answer (2)
NCERT Reference: Class XI, Part-II, Page No. 329

85. Answer (3)
NCERT Reference: Class XI, Part-II, Page No. 330

86. Answer (2)
NCERT Reference: Class XI, Part-II, Page No. 341

87. Answer (3)
NCERT Reference: Class XI, Part-II, Page No. 341

88. Answer (4)
NCERT Reference: Class XI, Part-II, Page No. 344

89. Answer (3)
NCERT Reference: Class XI, Part-II, Page No. 354

90. Answer (4)
NCERT Reference: Class XI, Part-II, Page No. 349

91. Answer (2)
NCERT Reference: Class XI, Part-II, Page No. 364

92. Answer (1)
NCERT Reference: Class XI, Part-II, Page No. 338

93. Answer (2)
NCERT Reference: Class XI, Part-II, Page No. 366

94. Answer (2)
NCERT Reference: Class XI, Part-II, Page No. 348

95. Answer (3)
NCERT Reference: Class XI, Part-II, Page No. 339

96. Answer (2)
NCERT Reference: Class XI, Part-II, Page No. 337

97. Answer (2)
NCERT Reference: Class XI, Part-II, Page No. 351

98. Answer (2)
NCERT Reference: Class XI, Part-II, Page No. 348

99. Answer (3)
NCERT Reference: Class XI, Part-II, Page No. 358

100. Answer (4)
NCERT Reference: Class XI, Part-II, Page No. 363

BOTANY

101. Answer (2)
NCERT Reference: Class XI, Page No. 194

102. Answer (4)
NCERT Reference: Class XI, Page No. 195

103. Answer (3)
NCERT Reference: Class XI, Page No. 197,198

104. Answer (3)
NCERT Reference: Class XI, Page No. 199

105. Answer (2)
NCERT Reference: Class XI, Page No. 199
106. Answer (1)
NCERT Reference: Class XI, Page No. 209
107. Answer (2)
NCERT Reference: Class XI, Page No. 210
108. Answer (4)
NCERT Reference: Class XI, Page No. 210
109. Answer (2)
NCERT Reference: Class XI, Page No. 211
110. Answer (3)
NCERT Reference: Class XI, Page No. 212
111. Answer (2)
NCERT Reference: Class XI, Page No. 213
112. Answer (1)
NCERT Reference: Class XI, Page No. 215
113. Answer (3)
NCERT Reference: Class XI, Page No. 216
114. Answer (4)
NCERT Reference: Class XI, Page No. 219
115. Answer (1)
NCERT Reference: Class XI, Page No. 220
116. Answer (4)
NCERT Reference: Class XI, Page No. 227,228
117. Answer (2)
NCERT Reference: Class XI, Page No. 229
118. Answer (3)
NCERT Reference: Class XI, Page No. 232
119. Answer (1)
NCERT Reference: Class XI, Page No. 229,232
120. Answer (2)
NCERT Reference: Class XI, Page No. 235
121. Answer (4)
NCERT Reference: Class XI, Page No. 240
122. Answer (3)
NCERT Reference: Class XI, Page No. 241
123. Answer (4)
NCERT Reference: Class XI, Page No. 243
124. Answer (4)
NCERT Reference: Class XI, Page No. 246
125. Answer (2)
NCERT Reference: Class XI, Page No. 247
126. Answer (1)
NCERT Reference: Class XI, Page No. 250
127. Answer (2)
NCERT Reference: Class XI, Page No. 252,253
128. Answer (2)
NCERT Reference: Class XI, Page No. 202
129. Answer (1)
NCERT Reference: Class XI, Page No. 223
130. Answer (2)
NCERT Reference: Class XI, Page No. 176
131. Answer (3)
NCERT Reference: Class XI, Page No. 178
132. Answer (3)
NCERT Reference: Class XI, Page No. 186,187
133. Answer (4)
NCERT Reference: Class XI, Page No. 185
134. Answer (2)
NCERT Reference: Class XI, Page No. 180
135. Answer (4)
NCERT Reference: Class XI, Page No. 189
136. Answer (4)
NCERT Reference: Class XI, Page No. 201
137. Answer (1)
NCERT Reference: Class XI, Page No. 203
138. Answer (3)
NCERT Reference: Class XI, Page No. 214, 215
139. Answer (3)
NCERT Reference: Class XI, Page No. 218
140. Answer (1)
NCERT Reference: Class XI, Page No. 223
141. Answer (3)
NCERT Reference: Class XI, Page No. 229
142. Answer (4)
NCERT Reference: Class XI, Page No. 233
143. Answer (1)
NCERT Reference: Class XI, Page No. 237
144. Answer (2)
NCERT Reference: Class XI, Page No. 245

145. Answer (1)

NCERT Reference: Class XI, Page No. 250

146. Answer (3)

NCERT Reference: Class XI, Page No. 230

147. Answer (4)

NCERT Reference: Class XI, Page No. 249

148. Answer (1)

NCERT Reference: Class XI, Page No. 177

149. Answer (1)

NCERT Reference: Class XI, Page No. 187

150. Answer (2)

NCERT Reference: Class XI, Page No. 189

ZOOLOGY

151. Answer (4)

NCERT Reference: Class XI, Page No. 50,51

152. Answer (3)

NCERT Reference: Class XI, Page No. 57,59,60

153. Answer (2)

NCERT Reference: Class XI, Page No. 48,52

154. Answer (4)

NCERT Reference: Class XI, Page No. 58,113

155. Answer (4)

NCERT Reference: Class XI, Page No. 111,114, 115

156. Answer (2)

NCERT Reference: Class XI, Page No. 50

157. Answer (4)

NCERT Reference: Class XI, Page No. 52

158. Answer (4)

NCERT Reference: Class XI, Page No. 56,58

159. Answer (2)

NCERT Reference: Class XI, Page No. 60

160. Answer (3)

NCERT Reference: Class XI, Page No. 112

161. Answer (2)

NCERT Reference: Class XI, Page No. 112

162. Answer (3)

NCERT Reference: Class XI, Page No. 114,115

163. Answer (4)

NCERT Reference: Class XI, Page No. 54

164. Answer (1)

NCERT Reference: Class XI, Page No. 50,51,54

165. Answer (1)

NCERT Reference: Class XI, Page No. 56

166. Answer (3)

NCERT Reference: Class XI, Page No. 57

167. Answer (2)

NCERT Reference: Class XI, Page No. 53

168. Answer (4)

NCERT Reference: Class XI, Page No. 53, 54

169. Answer (3)

NCERT Reference: Class XI, Page No. 48, 51

170. Answer (1)

NCERT Reference: Class XI, Page No. 56, 57

171. Answer (2)

NCERT Reference: Class XI, Page No. 50, 47

172. Answer (4)

NCERT Reference: Class XI, Page No. 114

173. Answer (4)

NCERT Reference: Class XI, Page No. 112,113, 114

174. Answer (1)

NCERT Reference: Class XI, Page No. 115

175. Answer (3)

NCERT Reference: Class XI, Page No. 112

176. Answer (4)

NCERT Reference: Class XI, Page No. 114

177. Answer (4)

NCERT Reference: Class XI, Page No. 57

178. Answer (3)

NCERT Reference: Class XI, Page No. 114,115

179. Answer (3)

NCERT Reference: Class XI, Page No. 53,106, 112

180. Answer (2)

NCERT Reference: Class XI, Page No. 113

181. Answer (3)

NCERT Reference: Class XI, Page No. 111

182. Answer (2)

NCERT Reference: Class XI, Page No. 47,51,60

183. Answer (3)
NCERT Reference: Class XI, Page No. 58,59

184. Answer (2)
NCERT Reference: Class XI, Page No. 52

185. Answer (3)
NCERT Reference: Class XI, Page No. 51

186. Answer (4)
NCERT Reference: Class XI, Page No. 57, 58

187. Answer (2)
NCERT Reference: Class XI, Page No. 116

188. Answer (3)
NCERT Reference: Class XI, Page No. 108,109,110

189. Answer (2)
NCERT Reference: Class XI, Page No. 118

190. Answer (3)
NCERT Reference: Class XI, Page No. 117

191. Answer (2)
NCERT Reference: Class XI, Page No. 108,109

192. Answer (1)
NCERT Reference: Class XI, Page No. 55,56,57

193. Answer (2)
NCERT Reference: Class XI, Page No. 55,116

194. Answer (2)
NCERT Reference: Class XI, Page No. 52,53

195. Answer (4)
NCERT Reference: Class XI, Page No. 58,59,60

196. Answer (3)
NCERT Reference: Class XI, Page No. 114

197. Answer (2)
NCERT Reference: Class XI, Page No. 116

198. Answer (4)
NCERT Reference: Class XI, Page No. 110

199. Answer (3)
NCERT Reference: Class XI, Page No. 120,53

200. Answer (4)
NCERT Reference: Class XI, Page No. 57,58,60

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